

ORIGINAL CONTRIBUTION

Efficacy of biorational insecticides against chilli thrips, *Scirtothrips dorsalis* (Thysanoptera: Thripidae), infesting roses under nursery conditions

L. F. Aristizábal¹, Y. Chen², R. H. Cherry³, R. D. Cave⁴ & S. P. Arthurs¹

¹ Mid-Florida Research and Education Center, University of Florida, Apopka, FL, USA

² Hammond Research Station, Louisiana State University Ag-Center, Hammond, LA, USA

³ Everglades Research and Education Center, University of Florida, Belle Glade, FL, USA

⁴ Indian River Research & Education Center, University of Florida, Fort Pierce, FL, USA

Keywords

azadirachtin, *Beauveria bassiana*,
Entomopathogenic fungi, horticultural oil,
Metarhizium brunneum, spinosad

Correspondence

Steven P. Arthurs (corresponding author),
Mid-Florida Research and Education Center,
University of Florida, Apopka, FL 32703, USA.
E-mail: spa@ufl.edu

Received: February 15, 2016; accepted:
May 24, 2016.

doi: 10.1111/jen.12340

Abstract

We evaluated biopesticides based on entomopathogenic fungi, azadirachtin and horticultural oils for the management of the chilli thrips, *Scirtothrips dorsalis* Hood, an economically significant invasive pest in the United States. Insecticides were applied four times at 7- to 10-day intervals against established *S. dorsalis* infestations on shrub roses KnockOut[®], *Rosa* x 'Radrazz' under simulated nursery conditions. When applied as stand-alone treatments, *Beauveria bassiana* GHA (BotaniGard[®] ES), *Metarhizium brunneum* F52 (Met-52 EC), a horticultural oil (SuffOil-X[®]) and azadirachtin (Molt-X[®]) at label rates provided significant control, reducing populations of *S. dorsalis* by 48–71% compared with control over 4–6 weeks. Similar results were observed when the biopesticides were applied in rotation with each other. A conventional standard, spinosad (Conserve[®] SC), was consistently the most effective treatment in these studies, reducing thrips populations by >95% overall. In another study, more effective control (87%–92%) was achieved in biopesticide rotation programmes that included spinosad, when compared with those that did not. Results also showed that these biopesticides can be tank-mixed. However, there was no evidence that *B. bassiana* or *M. brunneum* combined with azadirachtin resulted in additive or synergistic control, as neither tank-mix treatment improved control compared with azadirachtin alone. These findings highlight the potential use of biopesticides in rotation programmes with conventional insecticides to manage *S. dorsalis* on roses. Biopesticides evaluated in this study can be incorporated into an IPM programme for roses.

Introduction

Native to Asia, the chilli thrips, *Scirtothrips dorsalis* Hood (Thysanoptera: Thripidae), has emerged as an important pest of several economically important crops, including vegetables, fruits and ornamentals, in the south-eastern United States (Mound and Palmer 1981; Kumar et al. 2013). There is also evidence of host range shifts within its invasive range (Venette and Davis 2004; Klassen et al. 2008; Kumar et al. 2012). Both adults and larvae attack aboveground

portions of the plants, including leaves, buds and fruits, resulting in aesthetic and economic damage. In the south-eastern USA, *S. dorsalis* has become a principle pest of roses (Rosales: Rosaceae) including KnockOut[®] (*Rosa* x 'Radrazz') the most widely sold shrub rose series in North America (Hodges et al. 2005; Ring 2012; Mannion et al. 2013; Bogren 2015).

Chemical control is the primary management approach reported for *S. dorsalis*, with insecticides from several classes used in different crops and regions. Products containing organophosphates,

neonicotinoids, pyrethroids, spinosyns, avermectins, chlorfenapyr and tetroneic acid derivatives have been reported as having efficacy against *S. dorsalis* populations (Shibao 1997; Seal et al. 2006; Seal and Kumar 2010; Arthurs and Leibe 2011; Varghese and Mathew 2013; Naik et al. 2015). However, there are reports where excessive reliance on insecticides has resulted in resistance development. Resistance of *S. dorsalis* to older insecticides, including the organochlorine (DDT, BHC and endosulfan), carbamate (carbaryl), organophosphate (acephate, methyl-o-demeton, dimethoate, triazophos, monocrotophos) and pyrethroid classes, has been reported in its native range (Sridhar and Rani 2003; Vanisree et al. 2011; Yokoyama et al. 2014). Other thrips including the western flower thrips *Frankliniella occidentalis* Pergande and melon thrips *Thrips palmi* Karny have developed resistance to spinosad in the USA and elsewhere (Loughner et al. 2005; Gao et al. 2012a; Seal et al. 2013). Chemical control programmes for *S. dorsalis* may also suppress its natural enemies (Arthurs and Leibe 2011). Under the warm climate in the south-eastern USA, these risk factors are enhanced by a long growing season, which allows multiple thrips generations each year. Therefore, sole use of chemical insecticides to manage *S. dorsalis* is undesirable.

Biorational insecticides, including botanicals, horticultural oils and products based on entomopathogenic fungi, are alternatives or supplements to chemical insecticides. Products based on entomopathogenic fungi including *Beauveria bassiana* (Bals.-Criv.) Vuill., *Metarhizium brunneum* Petch and *Isaria fumosorosea* Wize have been tested against *F. occidentalis* and other thrips (Maniania et al. 2001, 2003; Ludwig and Oetting 2002; Ansari et al. 2008; Gao et al. 2012b). However, few reports are available on the use of entomopathogenic fungi against *S. dorsalis* (Arthurs et al. 2013). Azadirachtin, derived from neem seeds, *Azadirachta indica* (A. Juss), works as an insect growth regulator (ecdysone agonist), disrupting the moulting process. Azadirachtin may also act as a repellent, anti-feedant and oviposition-deterrent (Ascher 1993). Horticultural oils, which act by suffocation or cuticle disruption of soft-bodied arthropods, are another alternative available for thrips control. Considering the lack of studies in this area, here we evaluated entomopathogenic fungi, azadirachtin and refined horticultural oils against *S. dorsalis* infesting roses under simulated nursery conditions compared to spinosad as a conventional product. Treatments were compared in stand-alone, rotation and combination strategies to develop spray recommendations for nursery growers.

Material and Methods

Chilli thrips colony

Scirtothrips dorsalis collected from rose plants in Seminole County, Florida (28.63 N, 81.54W), were established as a colony in an insectary maintained at 25°C, 60% RH and 14 : 10 L : D photoperiod. Insects were maintained on 'Deltapine 493 Conventional' cotton plants (*Gossypium hirsutum* L.). Clean cotton plants were grown in a greenhouse in cages and added weekly to the chilli thrips colony.

Plants and experimental conditions

Studies were conducted in 2014 and 2015 at the Mid-Florida Research and Education Center in Central Florida. Rose plants 'KnockOut® Double Pink' were obtained from a local nursery and transplanted into 11.4-L containers. Test plants were pruned in March each year and every 3 months thereafter fertilized with Osmocote Plus 15N-3.9P-9.9K (Scotts-Sierra Horticultural Products Co., Marysville, OH) at 53g/container. The potting mix was 60% pine bark, 30% NuPeat (equal parts composted yard waste, hardwood bark and peat), and 10% coarse sand. Plants were held for 6 months prior to tests to ensure any prior insecticide residues had dissipated. Experimental blocks consisted of 12 modified shade houses, each 10 × 10 m, with all sides covered by a 3-m-high 70% shade netting to help isolate experimental blocks. The roof of each shade house was removed to allow full sun needed for rose cultivation. Plants were individually irrigated with microjet spray stakes (Irrigation Direct, Inc. Livermore, CA) connected via spaghetti tube and controlled via a central irrigation timer. Plants were watered daily at 2–4 l and adjusted based on the season. For all experiments, each replicate consisted of a group of three plants arranged at 45 cm spacing to simulate nursery conditions within each block, with a spacing of 3 m between treatments. Environmental data were obtained hourly from a weather station located adjacent to the field site.

Insecticides

Five biorational insecticides including two entomopathogenic fungi (mycoinsecticides), azadirachtin, a horticultural oil, and spinosad as the conventional standard were used the following experiments (table S1). All materials were applied within label rates; that is, *Beauveria bassiana* strain GHA at 2.5 ml/l (BotaniGard®ES containing 2.1×10^{13} spores/l;

Laverlam International Corp., Butte, MT), *Metarhizium brunneum* strain F52 at 2.5 ml/l (Met-52 EC containing 5.5×10^{12} spores/l; Novozymes Biologicals Inc., Salem, VA), azadirachtin at 0.6 ml/l (Molt-X[®], BioWorks Inc., Victor, NY), mineral oil at 10 ml/l (SuffOil-X[®], BioWorks Inc.) and spinosad at 0.63 ml/l (Conserve[®] SC, Dow Agro Sciences LLC, Indianapolis, IN). A surfactant (CapSil[®], Aquatrols, Paulsboro, NJ) was used at 1 ml/l in all treatments, except mineral oil. The viability of the mycoinsecticides before each test was checked by counting the number of conidia or spores that germinated after 24-h incubation on potato dextrose agar (2% w/v, Difco Laboratories, Detroit, MI) at 25°C, 12 : 12 L : D photoperiod. The germination was 85–94% for all entomopathogenic fungi tested, so products were considered viable.

Evaluation of insecticides as stand-alone treatments

All insecticides were tested as stand-alone treatments against chilli thrips populations on rose plants. A randomized complete block design was used for this experiment with six replicates per treatment. Plants were pruned by one-third of the stem length to stimulate new foliage growth. Two weeks later, plants were infested with 40 adult *S. dorsalis* following the methods described by Doğramaci et al. (2011). Three weeks later, when thrips populations increased to a damaging level and F1 progeny was observed, the first spray of four applications (at 7–10 day intervals) was applied at the stated label rates. Water with wetting agent alone was included to determine whether the surfactant impacted thrips populations.

Treatments were applied using a CO₂-pressurized sprayer (R&D Sprayers, Opelousas, LA) fitted with a cone nozzle. Both foliage and the upper soil surface were sprayed to target feeding and pupal stages, respectively. Total application volume was 86 ml/plant, equivalent to 1000L/ha. Applications were conducted in the late afternoon to minimize environmental stresses on the entomopathogenic fungi.

Evaluations were conducted 1–2 days before the first application and then weekly for 5 weeks. For population assessment, thrips adults and larvae were counted on 5 randomly selected terminals per plant using a 10× magnifier. The third leaf of each terminal was selected for assessing plant damage caused by thrips feeding injuries because this leaf was most likely to show symptoms when the plants were actively growing (L. F. Aristizábal, unpublished data). A damage rating at a 4-point scale was assigned, where 0 = no damage, 1 = slight (one scar), 2 = moderate (two to three scars or and deformations) and

3 = severe (more than three scars or severe deformation). The experiment was conducted in 2014 during summer (July to September) and repeated in fall (September to November). Environmental conditions during the summer trial remained generally hot and sunny (maximum 964 W/m²) with a total of 15.2 cm rainfall. Average daily minimum and maximum temperatures ranged from 21.1 to 37.1°C. Average daily relative humidity ranged from 69 to 93%. Conditions were cooler and wetter during the fall study, with average daily temperatures from 3.7 to 33.4°C, relative humidity from 59 to 95% and 17.6 cm of rainfall.

Efficacy of rotation programmes with biopesticides

Two insecticide rotation programmes were evaluated. Four spray applications were made at 7- to 10-day intervals in each programme. The first rotation programme compared the efficacy of biorational insecticides rotated with each other twice. The second rotation programme evaluated two entomopathogenic fungi, *B. bassiana* and *M. brunneum*, in rotation with spinosad. The objective for the latter programme was to evaluate the efficacy of rotation between entomopathogenic fungi and spinosad compared with spinosad alone. Our hypothesis was that substituting 50% (i.e. 2 of 4) applications of a standard conventional insecticide (spinosad) by entomopathogenic fungi would not impact the effectiveness of the programme. In both programmes, an untreated (no-spray) control and a conventional insecticide (spinosad) applied as a stand-alone were included. A water plus surfactant control was not included because no impact from CapSil was observed in the previous study. Chilli thrips infestation, insecticide application and data collection followed the procedures described previously. A RCBD was used for both rotation programmes, with eight and seven treatments, respectively, each with five replications. These two experiments were conducted simultaneously from May to July 2015. Conditions during the rotation experiments remained generally hot and sunny (maximum 957 W/m²) with a total of 32.6 cm rainfall. Average daily minimum and maximum temperatures ranged from 17.1 to 38.2°C. Relative humidity ranged from 69 to 91%.

Evaluation of tank mixtures

An additional study compared tank-mix treatments of *M. brunneum* and *B. bassiana*, each with azadirachtin, to stand-alone treatments of these three products. The goal was to assess the compatibility and potential

additive or synergistic effects between entomopathogenic fungi and azadirachtin, which has been noted elsewhere (Mohan et al. 2007; Islam et al. 2010; Hernández et al. 2012). Products were applied at label rates and combined in the spray tank. Spinosad and a no-spray control were included for comparison. Four applications were applied at 7- to 10-day intervals. Treatment effects were assessed weekly over 5 weeks following the same methods as the previous experiment. The study was conducted from July to August in 2015. Conditions remained generally hot and sunny (maximum 961 W/m²) with a total of 21.9 cm rainfall. Average minimum and maximum daily temperatures ranged from 21.2 to 36.4°C. Relative humidity ranged from 78 to 92%.

Data analysis

Treatment effects on the number of thrips were compared using repeated-measures ANOVA with between 5 and 7 time intervals (weeks). Mortality caused by

biopesticides was determined using Abbott's formula (Abbott 1925). Means were separated using Tukey's HSD tests in the repeated-measures model. All data were examined for normality and, if needed, normalized through $\log(n + 1)$ prior to analysis. The severity index (damage rating) at the end of each study was analysed via a nonparametric Kruskal–Wallis test. The analysis was performed in SPSS for Windows v.22 (IBM SPSS Statistics for Windows, Version 22.0., 2013; IBM Corp., Armonk, NY, USA).

Results

Evaluation of individual biopesticides

Thrips in the no-spray and water spray treatments reached ≈ 3.5 per sampled leaf in both summer and fall (fig. 1a,b). In contrast, roses treated with the four biopesticides reached ≈ 2.3 and 1.5 per leaf during the summer and fall trials, respectively. In the summer trial, both adults ($F_{6,35} = 33.0$, $P < 0.0001$) and larvae

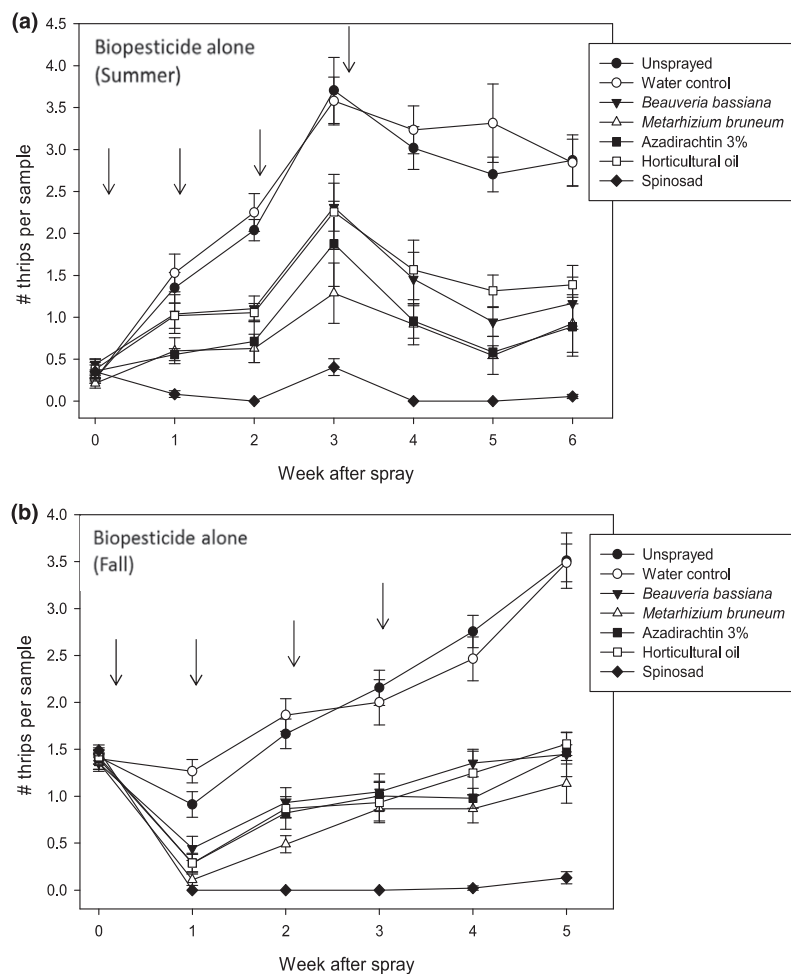


Fig. 1 Efficacy of biopesticides applied stand-alone against *S. dorsalis* on rose plants expressed as number of thrips per sampled leaf (mean $\pm$ SEM). Trials were conducted in summer (a) and fall (b) in 2014. Arrows indicate timing of treatment application.

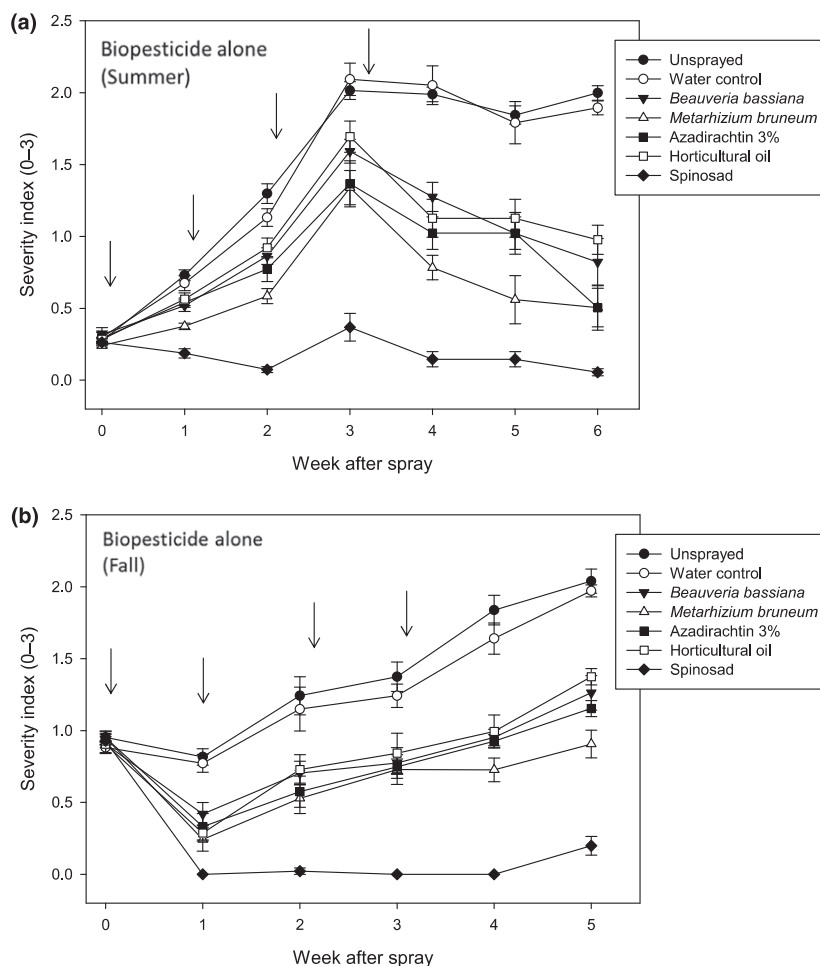


Fig. 2 Efficacy of biopesticides applied stand-alone against *S. dorsalis* on rose plants expressed as plant damage severity index on a 4-point scale (mean $\pm$ SEM). Trials were conducted in summer (a) and fall (b) in 2014. Arrows indicate timing of treatment application.

($F_{6,35} = 42.4$, $P < 0.0001$) were significantly affected by treatments (data not shown). Treatment effects of biopesticides were similar during the fall trial for adults ($F_{6,28} = 41.9$, $P < 0.0001$) and for larvae ($F_{6,28} = 49.7$, $P < 0.0001$). A comparison of means showed that spinosad was the most effective treatment in both experiments ($P < 0.05$, Tukey's HSD tests in the repeated model). There were no statistical differences among the four biopesticides in both experiments. There was also no difference between water/CapSil spray and the no-spray controls, suggesting that the surfactant had no effect on thrips. Using Abbott's formula, the reduction in *S. dorsalis* population by biopesticides with respect to the control over 6–7 weeks during summer and fall trials, respectively, was as follows: spinosad (97 and 99%), *M. brunneum* (71 and 72%), azadirachtin (61 and 67%), *B. bassiana* (47 and 55%) and horticultural oil (48 and 59%).

Severity index exhibited similar trends as thrips populations in both summer and fall experiment (fig. 2a,b). Low damage was observed on plants

treated with biopesticides compared with the no-spray or water spray treatments at week 2 or week 1 after treatment in the summer and fall experiment, respectively. Least damage was observed on plants treated with spinosad in both experiments. During the summer trial, severity index on biopesticide-treated plants decreased between weeks 3 and 6 due to more rapid plant growth and declining thrips populations. Analysis of plant damage (severity index) at the end of the study showed significant treatment effects in both the summer (Kruskal–Wallis $\chi^2 = 34.1$, $df = 6$, $P < 0.001$) and fall trial (Kruskal–Wallis $\chi^2 = 30.0$, $df = 6$, $P < 0.001$).

Efficacy of a rotation programme using biopesticides only

All combinations of biopesticides applied in rotation reduced *S. dorsalis* populations compared with the no-spray treatment over 6 weeks, but were less effective than spinosad (fig. 3a). Thrips in control plants reached 4.3 per leaf at weeks 6, compared with 1.6

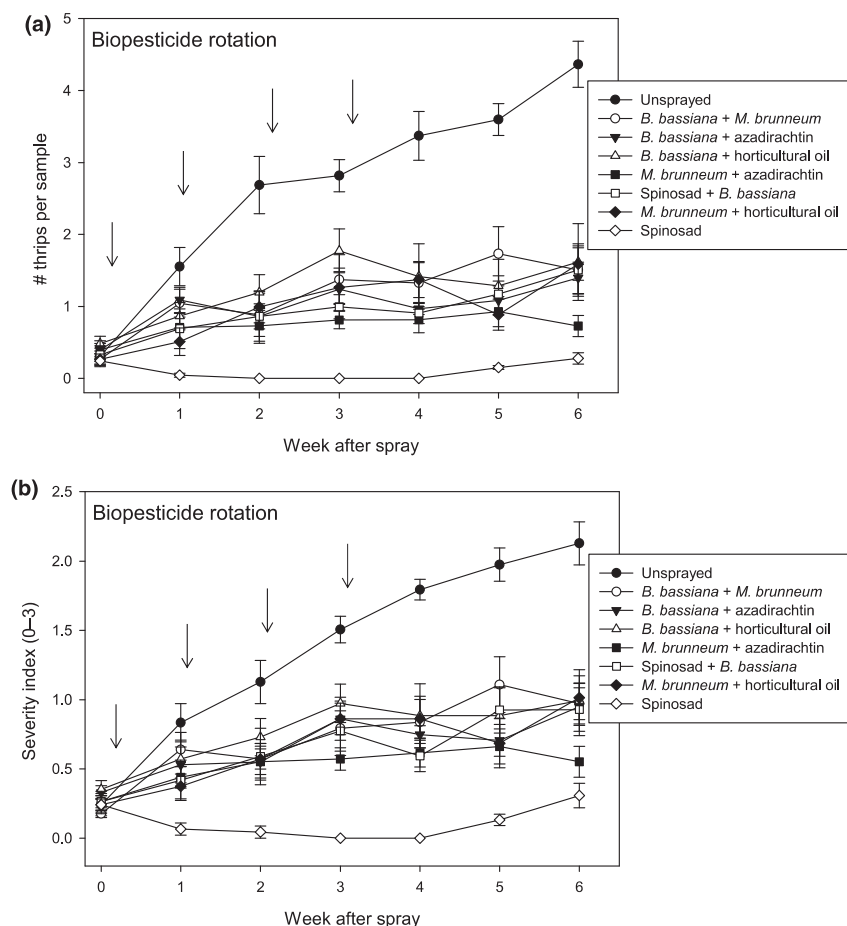


Fig. 3 Efficacy of biopesticides applied as rotation treatments with each other against *S. dorsalis* on rose plants expressed as number of thrips per leaf (a) and plant damage severity index on a 4-point scale (b) (mean $\pm$ SEM). Spinosad was included as a conventional standard. Arrows indicate timing of treatment application.

per leaf in the biopesticide rotation treatments. The effect of biopesticide treatments on thrips number was significant for both adults ($F_{7,32} = 12.9$, $P < 0.001$) and larvae ($F_{7,32} = 17.1$, $P < 0.001$). There were no significant differences among the various biopesticide treatments ($P > 0.05$, Tukey's HDS in the repeated measures model).

The rotation between *M. brunneum* and azadirachtin or the horticultural oil showed highest reduction in thrips population with 72.3% and 65.7% corrected reduction, respectively (fig 3a). Population reduction for other rotation treatments were as follows: azadirachtin + horticultural oil (63.9%), *B. bassiana* + azadirachtin (62.4%), *B. bassiana* + *M. brunneum* (52.7%) and *B. bassiana* + horticultural oil (53.7%). Spinosad had the highest effect against *S. dorsalis* throughout the experiment at 97.7% population reduction compared with no-spray control.

Severity index exhibited similar trends as thrips populations (fig. 3b). Severity index reached 2.1 in no-spray plants, while rotation treatments reached 1.0 and was lowest (0.3) for spinosad-treated plants. Severity index at week 6 indicated significant

treatment effects among no-spray, spinosad and biopesticide rotation treatments (Kruskal–Wallis $\chi^2 = 25.4$, $df = 7$, $P < 0.001$).

Efficacy of a rotation programme with mycoinsecticides and spinosad

All rotation programmes of mycoinsecticides with spinosad reduced *S. dorsalis* populations compared with the control treatment, which reached 4.4 individuals per leaf over 6 weeks (fig. 4a). Both adult and larval populations were significantly reduced by treatments (adult $F_{6,35} = 176.6$, $P < 0.0001$ and larvae $F_{6,35} = 81.5$, $P < 0.0001$). The rotation of *B. bassiana* + *M. brunneum* reduced thrips but was less effective compared with rotations of either mycoinsecticide with spinosad. All rotation treatments having spinosad as rotation partner maintained less than 1 thrips per leaf. There was no significant difference among rotation treatments that contained spinosad ($P > 0.05$, Tukey's HDS in the repeated measures model). Rotation treatments of spinosad with *M. brunneum* or *B. bassiana* resulted in 87.3 and 91.6% corrected population

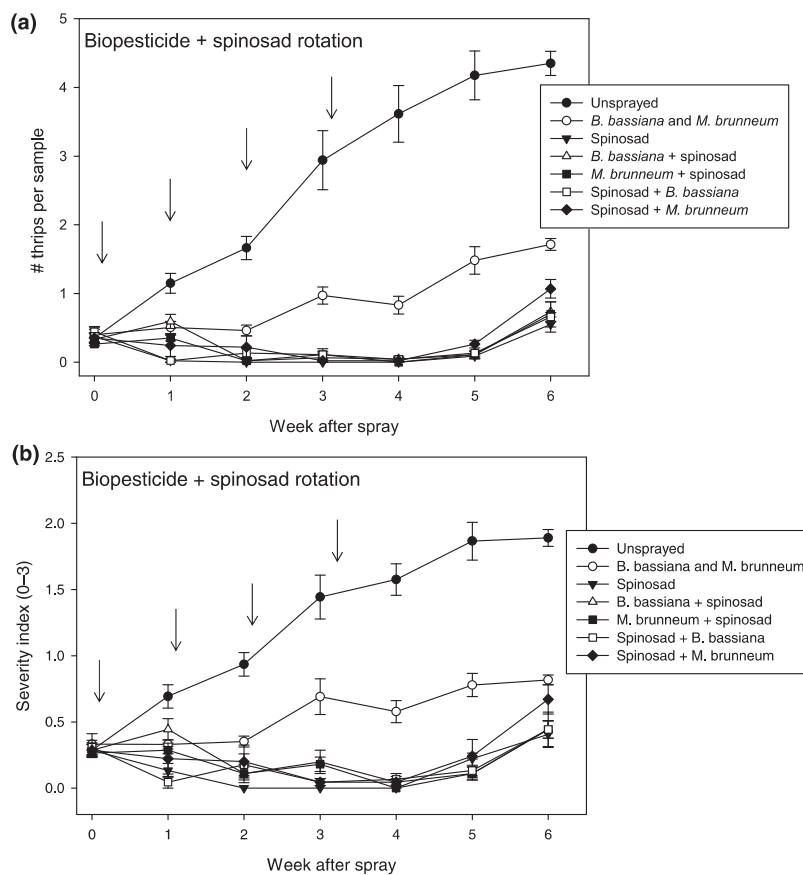


Fig. 4 Efficacy of two mycoinsecticides applied in rotation with each other or with spinosad against *S. dorsalis* on rose plants. Data show number of thrips per leaf (a) and plant damage severity index on a 4-point scale (b) (mean $\pm$ SEM). Arrows indicate timing of treatment application.

reduction, respectively. Corrected population reduction was 94.3% for spinosad alone and 65.1% for the *B. bassiana* + *M. brunneum* rotation.

Severity index was affected by treatments in a similar way as thrips populations. All treatments reduced damage severity compared with no-spray control (fig. 4b). Severity index at week 6 indicated significant treatment effects (Kruskal–Wallis $\chi^2 = 14.3$, df = 6, $P < 0.05$). Rotating either mycoinsecticide with spinosad or spinosad alone resulted in less plant damage than other treatments (fig. 4b).

Evaluation of tank mixtures of biopesticides

All treatments reduced thrips populations compared with the no-spray control, which reached 5.6 thrips per leaf by weeks 3 and 4 (fig. 5a). Both life stages were reduced by treatments (adult $F_{6,28} = 46.0$, $P < 0.0001$ and larvae $F_{6,28} = 50.9$, $P < 0.0001$). There was no evidence that *B. bassiana* or *M. brunneum* in tank mix with azadirachtin resulted in additive or synergistic control, as neither treatment resulted in better control than azadirachtin alone. The tank mixture of *M. brunneum* and azadirachtin caused an overall corrected thrips

mortality of 81.5%. Corrected mortality for other treatments was as follows *B. bassiana* and azadirachtin 74.3%, azadirachtin alone 72.9%, *M. brunneum* alone 72.1% and *B. bassiana* alone 57.1%. Spinosad was again the most effective treatment.

The expression of thrips feeding injuries, as indicated by severity index, was significantly lower in all insecticide treatments compared with the no-spray control (Kruskal–Wallis $\chi^2 = 29.2$, df = 6, $P < 0.0001$) (fig. 5b). Spinosad was the treatment with the lowest observed damage.

Discussion

Few reports are available on biopesticides for thrips management on ornamental plants. This research highlights the use of products containing two entomopathogenic fungi, a botanical extract (azadirachtin) and a horticultural oil for *S. dorsalis* management on roses grown under simulated nursery production conditions. Our data showed that products based on *B. bassiana*, *M. brunneum*, azadirachtin and a highly refined paraffin-based oil provided significant control, but were consistently less effective when compared

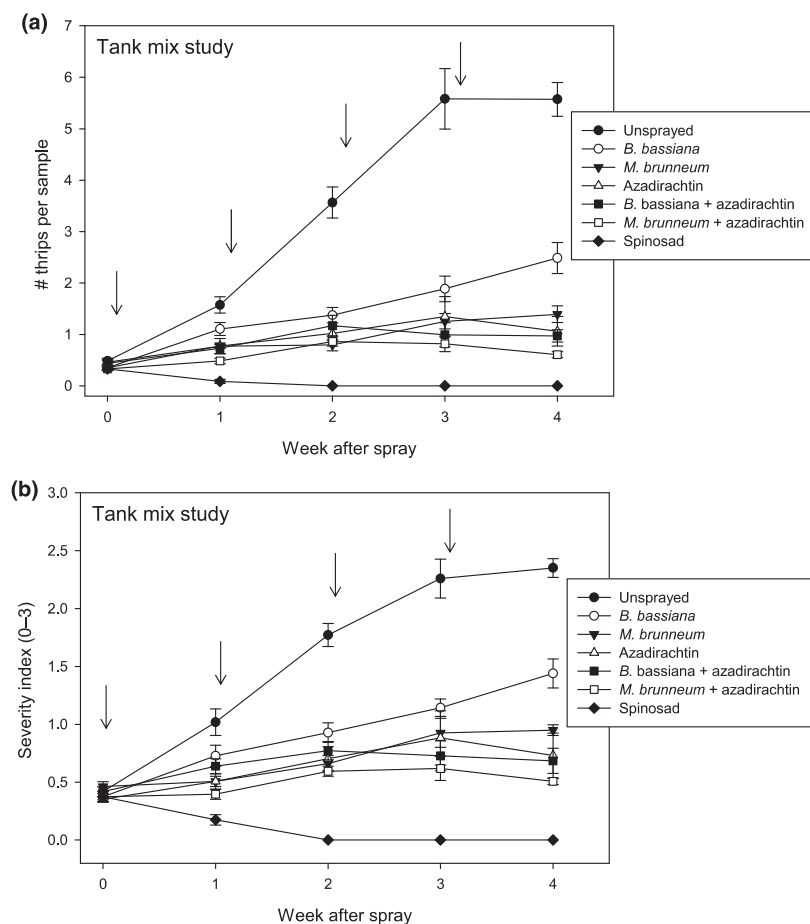


Fig. 5 Efficacy of two mycoinsecticides applied alone or as tank mixtures with azadirachtin against *S. dorsalis* on rose plants. Data show number of thrips per leaf (a) and plant damage severity index on a 4-point scale (b) (mean $\pm$ SEM). Arrows indicate the timing of treatment application.

with spinosad. Arthurs et al. (2013) tested entomopathogenic fungi as stand-alone treatments against *S. dorsalis* on roses and obtained similar results. This study also showed that biopesticides can be applied in rotation, but were less effective than spinosad. Another significant finding was that rotating spinosad with *B. bassiana* or *M. brunneum* did not significantly reduce efficacy compared to spinosad alone. This suggests that although less effective stand-alone, biopesticides can still be incorporated into a spray programme to reduce the use of and selection pressure on highly effective conventional products such as spinosad.

There are several reports of mycoinsecticides applied against *S. dorsalis* under greenhouse conditions. Seal and Kumar (2010) reported that *S. dorsalis* populations were reduced by $\approx 50\%$ at 5 DAT by *B. bassiana* applied at label rates to jalapeño pepper (*Capsicum annuum* L.). Higher efficacy was reported by Arthurs et al. (2013) who tested three applications of different mycoinsecticides applied against *S. dorsalis* on sweet pepper (*Capsicum annuum* L.). In these latter tests, *B. bassiana* achieved 81–94% mortality with

respect to controls, with *M. brunneum* providing 84–93%, and *Isaria fumosorosea* 62–66% control, respectively. Wu et al. (2013) screened 20 strains of *B. bassiana* for efficacy against *Thrips tabaci* (Linderman). The best strain (SZ-26) provided 83–100% mortality against adults at 10^7 conidia/mL and significantly reduced populations of both adults and larvae in greenhouse tests. These reports suggest that mycoinsecticides may work better under greenhouse conditions where environmental conditions can be manipulated to enhance infection.

Azadirachtin is derived from neem tree (*Azadirachta indica* A. Juss.) seeds and has been used commercially as an antifeedant and growth disruptor against various soft-bodied insects and mites (Ascher 1993). In our study, we expected that adding azadirachtin as a tank-mix partner would enhance the efficacy of entomopathogenic fungi for thrips control. It was hypothesized that an increased intermoult period for insect larvae by the growth regulating mode of action of azadirachtin may provide a better opportunity for the fungal conidia to contact and penetrate the insect

cuticle (Akbar et al. 2005). Previous studies investigating the interactive effects of entomopathogenic fungi and azadirachtin have reported variable results, with interactions including synergistic, additive and antagonistic (Akbar et al. 2005; Mohan et al. 2007; de Araujo et al. 2009; Islam et al. 2010). The reason that we did not observe additive or synergistic results in the present study is unclear. It may be that the residual activity of the fungal spores under the hot and sunny conditions observed during the test was too low to show a significant interaction with azadirachtin. Given that *S. dorsalis* primarily feeds on developing foliage and unopened flower buds, it is likely that high UV levels in the outer canopy were detrimental to the viability of fungal spores in these locations (Braga et al. 2001; Fernandes et al. 2007). It is also possible that the formulation of the azadirachtin tested (Molt-X[®]) was inhibitory to fungal germination, although this was not specifically tested.

The reduced risks of bioinsecticides on non-target organisms offer an advantage compared to conventional insecticides. The frequent use of conventional insecticides with high toxicity has caused concerns on their potential negative impacts, including outbreaks of phytophagous mites (Ako et al. 2004), negative effect on thrips predators (Arthurs and Leibe 2011), workers' safety and environmental impact. The preservation of natural enemies using less toxic biopesticides may enhance pest control. For example, Maniania et al. (2003) reported that *M. brunneum* applied at 10¹¹ conidia/ha resulted in a significant reduction of *T. tabaci* in an onion crop (*Allium cepa* L.). The reduced non-target effect preserved thrips natural enemies and was thought to allow *M. brunneum* treatments to be as effective as dimethoate (Maniania et al. 2003).

Nurseries producing ornamental plants have a low tolerance for insects which impact plant quality and marketability (Mizell and Short 1998). In our studies, thrips damage on roses treated with spinosad were low (severity index < 0.5), while plants treated with biopesticides showed slightly higher damage (severity index ranged between 0.4 and 1.6), and damage on the control plants were moderate to severe (severity index > 2). By themselves, the biopesticides tested are unlikely to provide sufficient control for growing roses under high population pressure of *S. dorsalis*. These biopesticides will need to be integrated into a spray programme with conventional products. It is a different scenario for managing *S. dorsalis* in roses planted in landscapes where tolerance for aesthetic damage is higher, elimination of the pest is not practical, and enhancing natural enemy activity is considered important (Dreistadt et al. 2004; Raupp et al.

2010). Therefore, rotating and tank mixing of biopesticides only may be an acceptable strategy for *S. dorsalis* management in the landscape.

In conclusion, research to optimize the use of biorational insecticides in an IPM programme for *S. dorsalis* in nursery rose production is presented. Further improvements in the use of biorational insecticides can be addressed through improved formulations, rotation schemes and integration with biological control agents in future research. Casey et al. (2007) provided an example of an IPM programme for cut-rose production in California using predatory mites for controlling *T. urticae*, and applications of azadirachtin and spinosad for controlling *F. occidentalis*. Similar strategies to include biorational insecticides need to be demonstrated to growers to encourage their use and adoption.

Acknowledgements

We are grateful to Ana Maria Bocsanzy, Robert Leckel, Moh Leng Kok-Yokomi and Celso Morales for technical support. The USDA-NIFA, Pest Management Alternatives (award 2013-34381-21203) and IR4 Programme provided financial support.

References

- Abbott WS, 1925. A method of computing the effectiveness of an insecticide. *J Econ Entomol*, 18, 265–267.
- Akbar N, Lord JC, Nechols JR, Loughin TM, 2005. Efficacy of *Beauveria bassiana* for red flour beetle when applied with plant essential oils or in mineral oil and organosilicone carriers. *J Econ Entomol*, 98, 683–688.
- Ako MC, Borgemeister HM, Poehling A, Elbert A, Nauen R, 2004. Effects of neonicotinoid insecticides on the biology of twospotted spider mite (Acari: Tetranychidae). *J Econ Entomol*, 97, 1587–1594.
- Ansari MA, Brownbridge M, Shad FA, Butt TM, 2008. Efficacy of entomopathogenic fungi against soil-dwelling life stages of western flower thrips, *Frankliniella occidentalis*, in plant-growing media. *Entomol Exp Appl*, 127, 80–87.
- Arthurs S, Leibe G, 2011. Insecticide trial for thrips control on plumbago, 2010. *Arthropod Manage Test*, 36 (G17).
- Arthurs SP, Aristizábal LF, Avery PB, 2013. Evaluation of entomopathogenic fungi against chilli thrips *Scirtothrips dorsalis*. *J Insect Sci*, 13, 31.
- de Araujo JM, Marques EJ, de Oliveira JV, 2009. Potencial de isolados de *Metarhizium anisopliae* e *Beauveria bassiana* e do óleo de nim no controle do pulgão *Lipaphis erysimi* (Kalt.) (Hemiptera: Aphididae). *Neotropical Entomol*, 38, 520–525.

- Ascher KRS, 1993. Nonconventional insecticidal effects of pesticides available from the neem tree, *Azadirachta indica*. Arch Inst Biochem Physiol, 22, 433–449.
- Bogren RC, 2015. Look out for chilli thrips this fall. Louisiana State University, Ag.Center. http://www.lsuagcenter.com/news_archive/2015/September/headline_news/look-out-for-chilli-thrips-this-fall.htm [accessed on January 2016].
- Braga GUL, Flint SD, Miller CD, Anderson AJ, Roberts DW, 2001. Variability in response to UV-B among species and strains of *Metarhizium* isolated from sites at latitudes from 61° N to 54° S. J Invertebr Pathol, 78, 98–108.
- Casey C, Newman J, Robb K, Tjosvold SA, MacDonald JD, Parrella MP, 2007. IPM program successful in California greenhouse cut roses. Calif Agr, 61, 71–78.
- Doğramaci M, Chen J, Arthurs PS, McKenzie CL, Irizarry F, Houben K, Brennan M, Osborne L, 2011. Mini-aspirator: a new device for collection and transfer of small arthropods to plants. Fla Entomol, 94, 22–27.
- Dreistadt SH, Clark JK, Flint ML, 2004. Pests of landscape trees and shrubs: an integrated pest management guide, vol. 3359. UCANR Publications, Oakland, CA.
- Fernandes EKK, Rangel DEN, Moraes AML, Bittencourt VR, Roberts DW, 2007. Variability and tolerance to UV-B radiation among *Beauveria* spp. isolates. J Invertebr Pathol, 96, 237–243.
- Gao Y, Lei Z, Reitz SR, 2012a. Western flower thrips resistance to insecticides: detection, mechanisms and management strategies. Pest Manage Sci, 68, 1111–1121.
- Gao Y, Reitz SR, Wang J, Xu X, Lei Z, 2012b. Potential of a strain of the entomopathogenic fungus *Beauveria bassiana* (Hypocreales: Cordycipitaceae) as a biological control agent against western flower thrips, *Frankliniella occidentalis* (Thysanoptera: Thripidae). Biocontrol Sci Technol, 22, 491–495.
- Hernández MM, Martínez-Villar E, Peace C, Pérez-Moreno I, Marco V, 2012. Compatibility of the entomopathogenic fungus *Beauveria bassiana* with flufenoxuron and azadirachtin against *Tetranychus urticae*. Exp Appl Acarol, 58, 395–405.
- Hodges G, Edwards GB, Dixon W, 2005. Chilli thrips *Scirtothrips dorsalis* Hood (Thysanoptera: Thripidae) a new pest thrips for Florida. Florida Department of Agriculture and Consumer Service, Department of Plant Industries, <http://www.freshfromflorida.com/Divisions-Offices/Plant-Industry/Plant-Industry-Publications/Pest-Alerts/Chilli-Thrips> [accessed on January 2016].
- Islam MT, Castle SJ, Ren S, 2010. Compatibility of the insect pathogenic fungus *Beauveria bassiana* with neem against sweet potato whitefly, *Bemisia tabaci*, on egg-plant. Entomol Exp Appl, 134, 28–34.
- Klassen W, Seal DR, Ciomperlik MA, Fieselmann DA, 2008. The chilli thrips, *Scirtothrips dorsalis*: current status in the Greater Caribbean Region. Proc Caribbean Food Crops Soc, 44, 103–117.
- Kumar V, Seal DR, Kakkar G, McKenzie CL, Osborne LS, 2012. New tropical fruit hosts of *Scirtothrips dorsalis* (Thysanoptera: Thripidae) and its relative abundance on them in south Florida. Fla Entomol, 95, 205–207.
- Kumar V, Kakkar G, McKenzie CL, Seal RD, Osborne LS, 2013. An overview of chilli thrips, *Scirtothrips dorsalis* (Thysanoptera: Thripidae), distribution and management. In: Agricultural and biological sciences weed and pest control - conventional and new challenges. Ed. by Soloneski S, Larramendy M, InTech Science Technology and Medicine open access publisher, 53–77.
- Loughner RL, Warnock DF, Cloyd RA, 2005. Resistance of greenhouse, laboratory, and native populations of western flower thrips to spinosad. Hort Sci, 40, 146–149.
- Ludwig SW, Oetting RD, 2002. Efficacy of *Beauveria bassiana* plus insect attractants for enhanced control of *Frankliniella occidentalis* (Thysanoptera: Thripidae). Fla Entomol, 85, 270–272.
- Maniania NK, Ekesi S, Löhr B, Mwangi F, 2001. Prospects for biological control of the western flower thrips, *Frankliniella occidentalis*, with the entomopathogenic fungus, *Metarhizium anisopliae*, on chrysanthemum. Mycopathologia, 155, 229–235.
- Maniania NK, Sithanatham S, Ekesi S, Ampong-Nyarko K, Baumgärtner J, Löhr B, Matoka CM, 2003. A field trial of the entomogenous fungus *Metarhizium anisopliae* for control of onion thrips, *Thrips tabaci*. Crop Prot, 22, 553–559.
- Mannion CM, Derksen AI, Seal DR, Osborne LS, Martin CG, 2013. Effect of rose cultivars and fertilization rates on population of *Scirtothrips dorsalis* (Thysanoptera: Thripidae) in Southern Florida. Fla Entomol, 96, 402–411.
- Mizell RF, Short DE, 1998. Integrated pest management in the commercial ornamental nursery. University of Florida/IFAS Extension Publication #ENY-336, <https://edis.ifas.ufl.edu/ig144> [accessed on February 2016].
- Mohan MC, Reddy NP, Devi UK, Kongara R, Sharma HC, 2007. Growth and insect assays of *Beauveria bassiana* with neem to test their compatibility and synergism. Biocontrol Sci Technol, 17, 1059–1069.
- Mound LA, Palmer JM, 1981. Identification, distribution and host plants of the pest species of *Scirtothrips* (Thysanoptera: Thripidae). Bull Entomol Res, 71, 467–479.
- Naik MN, Shetgar SS, Nalwandikar PK, Bhamare VK, 2015. Bio-efficacy of different insecticides against chilli thrips (*Scirtothrips dorsalis*). J Entomol Res, 39, 145–148.
- Raupp MJ, Shrewsbury PM, Herms DA, 2010. Ecology of herbivorous arthropods in urban landscapes. Ann Rev Entomol, 55, 19–38.
- Ring D, 2012. Chilli thrips threaten Louisiana knock out roses. Louisiana Agriculture Magazine, Winter 2012, p.

17. <https://www.lsuagcenter.com/NR/rdonlyres/4E5928A0-7956-40AD-B893-E88F3151069C/84964/LaAgMagwinter2012.pdf> (accessed on January 2016)
- Seal DR, Kumar V, 2010. Biological response of chilli thrips, *Scirtothrips dorsalis* Hood (Thysanoptera: Thripidae), to various regimes of chemical and biorational insecticides. *Crop Prot*, 29, 1241–1247.
- Seal DR, Ciomperlik M, Richards KL, Klassen W, 2006. Comparative effectiveness of chemical insecticides against the chilli thrips, *Scirtothrips dorsalis* Hood (Thysanoptera: Thripidae), on pepper and their compatibility with natural enemies. *Crop Prot*, 25, 949–955.
- Seal DR, Kumar V, Kakkar G, Mello SC, 2013. Abundance of adventive *Thrips palmi* (Thysanoptera: Thripidae) populations in Florida during the first sixteen years. *Fla Entomol*, 96, 789–796.
- Shibao M, 1997. Effects of insecticide application on population density of the chilli thrips, *Scirtothrips dorsalis* Hood (Thysanoptera: Thripidae), on grape. *Appl Entomol Zool*, 32, 512–514.
- Sridhar V, Rani BJ, 2003. Relative resistance in open and greenhouse populations of *Scirtothrips dorsalis* Hood (Thysanoptera: Thripidae) on rose to dimethoate and acephate. *Resist Pest Manage Newslett*, 12, 64.
- Vanisree PR, Upendhar S, Ramachandra Rao G, Srinivasa Rao V, 2011. Insecticide resistance in chilli thrips, *Scirtothrips dorsalis* (Hood) in Andhra Pradesh. *Resist Pest Manage Newslett*, 20, 22–25.
- Varghese TS, Mathew TB, 2013. Bioefficacy and safety evaluation of newer insecticides and acaricides against chilli thrips and mites. *J Trop Agr*, 51, 112–115.
- Venette RC, Davis EE, 2004. Chilli thrips/yellow thrips, *Scirtothrips dorsalis* hood (Thysanoptera: Thripidae) mini pest risk assessment. University of Minnesota, St. Paul, MN, USA.
- Wu S, Gao Y, Xu X, Zhang Y, Wang J, Lei Z, Smagghe G, 2013. Laboratory and greenhouse evaluation of a new entomopathogenic strain of *Beauveria bassiana* for control of the onion thrips *Thrips tabaci*. *Biocontrol Sci Technol*, 23, 794–802.
- Yokoyama T, Kashima T, Toda S, 2014. Point mutations of the sodium channel gene associated with the fenpropathrin resistance in *Scirtothrips dorsalis* and discrimination of resistant individuals using the PCR-RFLP assay. *Jpn J Appl Entomol Zool*, 58, 59–62.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Bioinsecticides applied as foliar sprays in tests against *Scirtothrips dorsalis* on roses in 2014/15.